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The Registered Wide-Binary Target under a Local a_0 : A Conditional Charge Meter, and the EFE-Present Exact Two-Body Solve

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Abstract

The framework built on $a_0 = \kappa c \sqrt{(G\rho_\Lambda)} = 9.3619 \times 10^{-11} \text{ m/s}^2$ carries a hash-frozen, pre-registered Gaia DR4 wide-binary target (Amendment 10: velocity-boost band $\gamma_v = 1.1614\text{--}1.1814$ canonical / $1.1917\text{--}1.2267$ alternate footing). This note reports two refereed results *about* that registration — the registration itself is untouched.

1. A conditional charge meter. In this framework a_0 is a local quantity — $\mathcal{A}(\mathcal{Q}) = a_0^2 \propto -(\mathcal{Q})$ depends on the local shift-charge density — and the operative (steady-state-continuity) reading of the charge a Milky-Way-class flow carries at the solar circle suppresses the local a_0 by an amount governed by the framework's one free dimensionless charge $\nu_0 \in [2.14 \times 10^{-5}, 1.77 \times 10^{-4}]$. Propagated to the registered asymptote: at the ν_0 **floor** the target moves ≤ 0.006 at every corner of the solar-overdensity bracket and both footings — inside the frozen bands' own widths. At the **ceiling** the move is corner-split: -0.004 (sparse corner) to -0.093 canonical / -0.109 alternate (dense corner, $\sim 5\times$ the band width). **DR4 therefore functions as a ν_0 -meter only conditionally** — decisive if the solar-circle charge overdensity sits toward the dense (abundance-matched, fast-assembly) corner, silent at the sparse corner. Settling that overdensity is the named prerequisite.

2. The EFE-present exact two-body solve. The registered band's top was built by applying an *EFE-free* exact two-body benchmark (finite- y factors 1.41–1.74) to the *EFE per-star* response. We solve the full nonlinear AQUAL pair in a uniform external field (colinear, axisymmetric; machinery gates: a $\mu \equiv 1$ run returns $B = 1.002$, and the small-mass limit returns 1.450 vs the closed-form linear 1.4732). Result: the exact pair boost is **0.71–0.99 $\times$** the linear per-star value at 5–30 kAU, rising monotonically toward it — the EFE-free benchmark correction **overcorrects with the external field on**. Both directions stated: the registered band top is *conservative* (a >1.23 no-verdict becomes less likely), **and** the same suppression puts the colinear per-pair signal $\sqrt{B} = 1.03\text{--}1.21$ at or below the frozen floor 1.1614 over most anchor separations — a DR4 reading *below* the registered band becomes more likely as well.

1. The frozen registration

The DR4 pre-registration (PREREGISTRATION_DR4.md, ten hash-stamped amendments) fixes, in force: $\gamma_v = 1.1614\text{--}1.1814$ (canonical footing) / $1.1917\text{--}1.2267$ (alternate), no-verdict edge 1.23, on the Route A kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$ with the external-field closure at the observed solar-neighbourhood $x_{\text{ext}} = 1.9$ (canonical a_0 units). The registered asymptotes reproduce here to the digit: $\sqrt{\nu(y(1.9))} = 1.2138$ (registered 1.2139), $\sqrt{\nu(y(1.577))} = 1.2586$ (registered 1.2592). Nothing in this note edits any frozen file; whether these results become Amendment 11 is a decision for the registration's amendment protocol, not made here.

2. Why the target could move at all: a_0 is local, and the drain reading is now settled

The framework's a_0 is set by the dark sector's pressure at the local charge density. A halo that *kept* its captured charge would suppress the local a_0 fatally (that branch is dead independently); a draining halo carries only the in-transit flow. Two readings of that flow's density coexisted in the corpus — a residence-time estimate and steady-state continuity, disagreeing by $16\text{--}119\times$. That fork is now settled: **for the committed picture (smooth capture, cold radial infall, no halt), mass conservation makes $\rho = / (4\pi r^2 v)$ exact; the residence estimate was the same argument with an inconsistent virialised shape.** Continuity is operative. Its solar-circle overdensity, re-derived from committed anchors both ways, brackets $1.5 \times 10^3\text{--}1.6 \times 10^4$ (mean-charge units) — the dense corner being abundance-matched halo mass with standard fast assembly, the sparse corner baryon-scaled mass accreted over a full Hubble time.

3. The conditional ν_0 -meter

Local suppression $S = a_{0,\text{loc}}/a_0(0) = [(1+\nu_0^2)/(1+\nu_0^2 r^2)]^{\frac{1}{4}}$ at solar-circle overdensity r raises the effective external-field parameter $x_{\text{ext}} \rightarrow x_{\text{ext}}/S$, and the registered asymptote follows the kernel closure. The full sweep (both footings, three bracket corners, both ν_0 edges):

footing	solar od	ν_0	S	γ	$\Delta\gamma$ vs banked
can	1.5×10^3	floor	0.9997	1.2137	-0.0001
can	1.5×10^3	ceiling	0.9831	1.2100	-0.0038
can	3.28×10^3	ceiling	0.9300	1.1980	-0.0157
can	1.47×10^4	floor	0.9767	1.2086	-0.0052
can	1.47×10^4	ceiling	0.5990	1.1209	-0.0928
alt	3.28×10^3	ceiling	0.9300	1.2403	-0.0183
alt	1.47×10^4	ceiling	0.5990	1.1500	-0.1085

Reading rules, stated symmetrically: γ inside the frozen band $\Rightarrow \nu_0$ near its floor *or* the sparse corner is real (degenerate — the meter is silent); $\gamma \approx 1.12\text{--}1.15$ (can) / $1.15\text{--}1.19$ (alt) $\Rightarrow$ ceiling charge *and* dense corner (the meter fires); γ at neither $\Rightarrow$ the continuity reading itself is charged. The degeneracy breaks only by independently settling the solar-circle charge overdensity — a local, in-principle-measurable quantity. That is the named prerequisite for reading DR4 as a charge meter.

4. The EFE-present exact two-body solve

The linearized AQUAL operator about the external background, $\mu_0[\nabla^2 + L_0(\hat{e} \cdot \nabla)^2]\varphi = 4\pi G\rho$ ($\mu_0 = 1/\nu_0$, $L_0 = d \ln \mu / d \ln x = 0.3674$ canonical), has a closed-form anisotropic Green's function; the pair force is exact at $O(\varepsilon)$: $B(\theta) = \nu_0 / \sqrt{1 + L_0 \sin^2 \theta}$, spanning 1.2598 () to 1.4732 () — a 17% orientation spread. The full nonlinear colinear solve (cylindrical grid, Route A kernel, observed-x boundary condition with the linear pair far field) is gated by two machinery checks — $\mu \equiv 1$ returns $B = 1.002$; the small-mass limit returns 1.450 vs 1.4732 — and gives, for equal solar masses:

s [kAU]	y_int	B_exact	B_exact/B_	$\sqrt{B_exact}$
5	2.53	1.051	0.713	1.025
10	0.63	1.183	0.803	1.088
20	0.16	1.345	0.913	1.160
30	0.07	1.454	0.987	1.206

The exact pair sits *below* the linear per-star value everywhere and approaches it monotonically from below — the external field suppresses the mutual deep-MOND enhancement that the EFE-free benchmark (finite-y factors 1.41–1.74) was built from. Consequences, both directions: the registered band **top** (which applied that benchmark multiplicatively) is conservative, so a >1.23 no-verdict is *less* likely; and the per-pair $\sqrt{B}$ sits at or below the registered **floor** 1.1614 for $s \leq 20$ kAU, so a DR4 reading below the band is *more* likely. Box-size, grid, and boundary sensitivities are 0.2–1.6% (checked); the perpendicular nonlinear case needs 3D and is carried as the stage's stated approximation (its *linear* value is exact above); the alternate-footing tensor ($\nu_0 = 1.5840$, $L_0 = 0.4112$) is on record, the nonlinear table being canonical-footing.

5. What this does not establish

- These are **asymptote-level and per-pair numbers**, not the composed band observable: the registration's composition grades (population synthesis, orientation statistics, mass spectrum) shift both results by the same benchmark factors as the amendment itself. The *directions* are the findings.
- The ν_0 -meter is **conditional on the solar-overdensity corner**, and its favourable reading (a new instrument) rests on the dense corner; the sparse corner shows nothing. Both are in the table.
- The two-body result is **colinear-exact only**; the perpendicular nonlinear correction is plausibly also ≤ 1 (the suppression mechanism is orientation-blind) but unproven here.
- Nothing here modifies the frozen registration, its hash chain, or the in-force band.

Reproducibility

All claims are reproduced by committed scripts at <https://github.com/carlzimmerman/zimmerman-formula>:

- `nbody_2026/stage61_drain_reading_fork_2026.py` — the drain-reading settlement and the conditional ν_0 -meter sweep (10 checks).
- `nbody_2026/stage64_efe_two_body_exact_2026.py` — the EFE-present exact two-body solve with both machinery gates and both-direction consequences (10 checks).

- `prep_2026/gaia_dr4_prep/aqual_efe_full_solve_2026.py` — the Amendment-10 full solve this note’s linear theory is pinned against (18 checks).
- `prep_2026/gaia_dr4_prep/PREREGISTRATION_DR4.md + AMENDMENT*_HASH.txt` — the frozen registration (untouched; verify the hashes).

Prior art and attribution. AeST is Skordis & Złośnik, PRL **127** 161302 (arXiv:2007.00082); the kernel is Milgrom & Sanders 2008, ApJ **678** 131, Eq. (13) at $\alpha = \frac{1}{2}$, adopted by McGaugh, Lelli & Schombert 2016. The anisotropic linearized-EFE response tensor follows the standard AQUAL linearization; the framework’s contribution is the a_0 normalisation, the local- a_0 structure, and the pre-registered target this note stress-tests.